Prism Gate Diagonal Baffle Photon Reader

Mathematical analysis and experimental program

Concept originator: Levi Burd
Technical development: ChatGPT (V001 to V003)
V004 revision and independent cross-check: Claude (Anthropic)

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Abstract

This paper evaluates a six-inch slide-in collection frame, a gated diagonal-mirror chamber, readers coupled through or integrated with its walls, and controlled release of the outgoing optical signal. It develops a dimensioned example, ray and wave propagation, timing constraints, loss and energy budgets, classical taps, nondestructive presence detection, and readout of a specified quantum observable. The full capture, reading, and release operation is included in the mathematical model.

The resulting architecture supports a testable optical buffer and sensing experiment. A reader behind a mirror needs a physical coupling channel. A nondestructive reader may register photon presence or measure a chosen observable under specified conditions. Deterministic acquisition of distinguishing information about nonorthogonal input states while preserving each state exactly is excluded by the joint evolution derived here. Repeated passes improve some readouts while accumulating loss or disturbance; they do not remove that constraint.

Evidence and intended use

The geometry and numerical cases are proposed engineering examples. The equations are analytical models with stated assumptions, and the accompanying programs verify numerical identities and representative state evolution. No device was built, no material interaction strength was measured, and no deployed link was tested. This paper is suitable for evaluating a bench research proposal; it is not a fabrication release, a certification, or evidence of recovered encryption keys.

Version 4.0 incorporates the internal technical review of V002 (Appendix C) and an independent cross-check of V003 by a second model that did not write the earlier versions (Appendix D). Established physical principles are identified as such; novelty of the assembly has not been assessed.

Key results and evidence status

Table 1. Each headline claim and the kind of evidence behind it.

Result	Basis	Status
80 mm square ring fits a 152.4 mm frame with 8.43 mm radial allowance	Geometry, Section 2	Computed
Bare loop transit 1.0674 ns; switching must finish within one transit	Equations 10 and 11	Computed
1.79 percent survival after 10 000 circuits even at $R = 0.9999$ and $T = 1$	Equation 12, Table 4	Computed, optimistic
Exact preservation of nonorthogonal states forbids any distinguishing record	Equations 30 to 32	Proved under stated assumptions
Basis reading trades record quality against complementary-basis error	Equation 35, Table 5	Proved for the stated probe model
Nondestructive presence detection exists in atom-cavity systems	References [1], [2]	Cited; not shown for this device
A wall reader with a specified coupling and loss budget	Findings R1, R6	Open; blocks fabrication
A stable transverse mode for long storage	Section 3	Open engineering deliverable

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1 Scope and mathematical state

Light is the electromagnetic field; photons are its quantized excitations. Detecting light can reveal arrival times, intensity, spectra, or an encoded observable. These outcomes require different instruments. A surviving photon and an unchanged encoded quantum state are separate experimental claims.

For a normalized single-photon wave packet with two polarization components, a useful input model is

$$|\Psi\rangle = \sum_{\sigma\in\{H,V\}} \int_0^\infty d\omega\, f_\sigma(\omega)\, a_\sigma^\dagger(\omega)|\mathrm{vac}\rangle, \qquad \sum_\sigma \int_0^\infty |f_\sigma(\omega)|^2 d\omega = 1. \tag{1}$$

If polarization separates from the other degrees of freedom, the polarization state is a qubit. Failure of that separation can itself cause decoherence when frequency or spatial modes are ignored.

$$|\psi\rangle = \alpha|H\rangle + \beta|V\rangle, \qquad |\alpha|^2 + |\beta|^2 = 1. \tag{2}$$

Let P denote the signal and M the entire reader, including probes, wall excitations, recorded outcomes, control memory, and inaccessible environments. The device induces a channel on the joint system. A lost photon must remain in the outcome space rather than disappear from the accounting.

$$\rho'_{PM} = U(\rho_P \otimes \rho_M)U^\dagger, \qquad \rho'_P = \mathrm{Tr}_M\, \rho'_{PM}. \tag{3}$$

Working definitions

Table 2. Principal symbols. Appendix E lists every symbol used in the paper.

Symbol	Meaning
N, n_{mir}	Number of circuits; mirror stations per circuit
K, r	Total reader encounters; reader encounters per circuit, $K = rN$
L_{rt}, R_j	Round-trip optical path; power reflectivity of mirror j
P_{out}	Unconditional probability of an accepted outgoing photon
F_{cond}	State fidelity conditional on an accepted surviving event
Y, b	Recorded reader outcome; a prepared input bit
Q_X, Q_Z	Errors measured in the complementary X and Z bases

The models below assume standard quantum mechanics. Orthogonal known-basis signals, unknown states drawn from a nonorthogonal alphabet, and optical side channels are treated separately. The completed analysis does not require assuming that every intercepted photon necessarily raises an alarm.

2 Mechanical and optical architecture

The proposed front frame has nominal outside diameter 152.4 mm. Its central opening admits the beam; there is no fabric or generic particle-reading screen. A clear diameter of 130 mm is

an illustrative mechanical allocation, subject to mount and switch dimensions. Input and output telescopes, electronics, and a quantum interaction module may extend beyond the frame; the frame diameter is not a demonstrated total package size.

A mode-matching stage routes the admitted pulse into a four-station square ring. The nominal mirror centers are at the following coordinates in metres, in the ring plane:

$$\mathbf{p}_0 = (-0.04, -0.04), \quad \mathbf{p}_1 = (0.04, -0.04), \quad \mathbf{p}_2 = (0.04, 0.04), \quad \mathbf{p}_3 = (-0.04, 0.04). \tag{4}$$

$$L_{\mathrm{s}} = 0.080\ \mathrm{m}, \qquad L_{\mathrm{rt}} = 4L_{\mathrm{s}} = 0.320\ \mathrm{m}, \qquad D_{\mathrm{c}} = L_{\mathrm{s}}\sqrt{2} = 0.11314\ \mathrm{m}. \tag{5}$$

The mirror-center circle, of diameter D_{c}, leaves about 8.43 mm radial clearance inside the illustrative 130 mm opening. That clearance must accommodate footprints, mounts, and tolerances; it is not a collision-free CAD result. Finite mirror dimensions can exceed this allocation. The optical switch is drawn on one segment and requires a three-dimensional mechanical design.

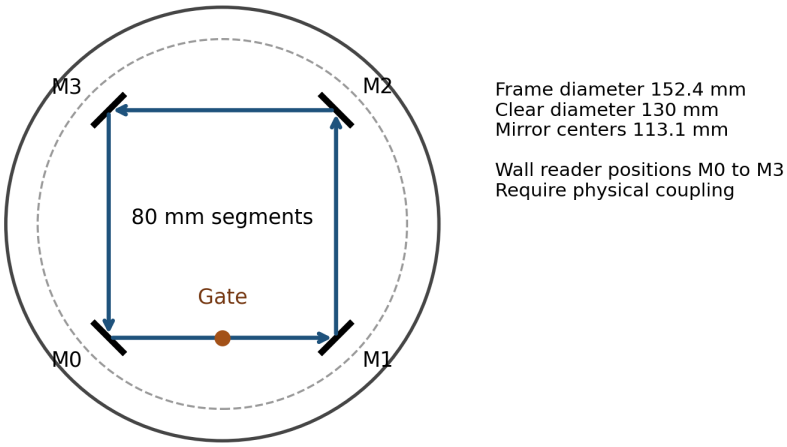


Figure 1. Nominal ring centerline and frame envelope. Reader ports at the walls are functional locations, not a validated microscopic interaction design.

3 Reflection and beam propagation

At each station let $\mathbf{d}$ be a unit propagation direction and $\mathbf{n}$ a unit surface normal. Ideal specular reflection obeys

$$\mathbf{d}_{\mathrm{out}} = \mathbf{d}_{\mathrm{in}} - 2(\mathbf{d}_{\mathrm{in}} \cdot \mathbf{n})\mathbf{n}, \qquad \mathbf{n} \parallel \mathbf{d}_{\mathrm{in}} - \mathbf{d}_{\mathrm{out}}. \tag{6}$$

These normals turn each square-ring segment through 90 degrees, giving 45-degree incidence. A true layout must trace both chief rays and beam envelopes through baffles, switches, mounts, and apertures. A folded ray path alone does not establish long-term transverse confinement.

For a paraxial Gaussian beam, a useful first check in free propagation is

$$z_R = \frac{\pi w_0^2}{\lambda}, \qquad w(z) = w_0\sqrt{1 + (z/z_R)^2}, \qquad \theta_{\mathrm{div}} \simeq \frac{\lambda}{\pi w_0}. \tag{7}$$

At an illustrative wavelength of 850 nm and waist radius of 0.50 mm, the Rayleigh length is 0.924 m and divergence is 0.541 mrad. The wavelength is an example, not an identification of the pictured link. The mirror footprint expands along the incidence plane by approximately the inverse cosine of the incidence angle θ_i.

$$w_{\text{footprint}} \simeq \frac{w}{\cos\theta_i}, \qquad \epsilon_{\text{clip}} = \exp(-2r_a^2/w^2). \tag{8}$$

The clipping expression applies to a centered circular aperture of radius r_a and a circular Gaussian intensity distribution. Tilted elliptical footprints and displaced beams require an appropriate two-dimensional integral. For a centered aperture radius equal to three beam radii, the ideal clipped fraction is about 1.52×10^{-8}.

Longer storage

A ring made only of plane mirrors is not a robust stable Gaussian resonator. Focusing elements or curved mirrors must be designed, including separate sagittal and tangential behavior at oblique incidence. For a real, lossless paraxial round-trip ray-transfer matrix $\mathcal{M}$ with determinant one, the stability and mode conditions are

$$\mathcal{M} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \qquad \left|\frac{A+D}{2}\right| < 1, \qquad q = \frac{Aq+B}{Cq+D}, \quad \operatorname{Im} q > 0. \tag{9}$$

The present coordinates specify the chief-ray geometry only. A selected stable cavity, surface curvatures, aperture analysis, and wave-optics tolerance model remain engineering deliverables before procurement.

4 Admission, storage, and release timing

The switched-loop schedule considered here uses a scheduled pulse or a separate herald; other coherent loading schemes are outside this design. An ordinary detector cannot absorb an incoming photon to announce its arrival and then release that same photon intact. A gate routes the pulse into the loop, changes state before it returns, and later routes it to the output.

$$\begin{aligned} \tau_{\text{rt}} &= \frac{1}{c}\sum_i n_{g,i}L_i + \tau_{\text{modules}}, \quad \tau_{\text{modules}} = \left.\frac{d\phi_{\text{modules}}}{d\omega}\right|_{\omega_0}, \\ \tau_{\text{hold}} &= N\tau_{\text{rt}}, \qquad \Delta\tau = \tau_{\text{fixed}} + \tau_{\text{hold}} - \tau_{\text{baseline}}. \end{aligned} \tag{10}$$

For the bare 0.320 m all-air path, neglecting component phase delays, one circuit lasts 1.0674 ns. Resonant readers and switches can add delay or reshape the pulse. The module term uses the transfer-phase derivative with the Fourier convention $e^{-i\omega t}$; it is a narrowband group-delay model. A 1000-circuit bare-path hold gives 1.067 μs, not a prediction for an integrated reader.

$$t_{\text{pulse}} + t_{\text{transition}} + t_{\text{guard}} < \tau_{\text{rt}}. \tag{11}$$

This inequality is a conservative switching condition for the proposed routing schedule, not a universal theorem for all optical buffers. A pulse longer than the loop transit can overlap its own

return at the gate. Ordinary slow shutters cannot implement this compact-loop schedule. A longer optical delay, shorter synchronized pulses, or a different coupling architecture is required.

Control states

Table 3. Control sequence for one trial.

State	Action and record
Align	Use a separate reference to characterize the path and pointing
Arm	Prepare the reader and synchronize the capture window
Admit	Route the scheduled pulse into the circulating mode
Hold and sense	Count scheduled circuits and apply the selected interaction
Release	Route the next selected circuit to the outgoing mode
Reset	Reprepare probe and gates for the next trial

Gate-command timestamps are classical control records. They do not establish photon occupancy unless a separate sensing interaction supplies that evidence. A herald has its own efficiency and false-trigger statistics, which must be included in trial selection.

Storage creates delay. Routing a shorter alternative path may offset some insertion delay relative to a particular baseline, but an instrument cannot release a photon before it arrives. Whether a receiver resolves the change depends on its timing windows and statistical measurements. Undetectability cannot be inferred from a small physical enclosure.

5 Throughput, energy, and coherence budgets

Let the fixed input and output coupling product be η. With n_{mir} mirrors per circuit and nonmirror circuit transmission T, a first-order independent-loss model is

$$P_{\mathrm{out}}(N) = \eta \left(T \prod_{j=1}^{n_{\mathrm{mir}}} R_j \right)^N, \qquad \eta = \eta_{\mathrm{in}} \eta_{\mathrm{out}}. \tag{12}$$

R_j and T are power or single-photon survival probabilities, not field amplitudes. T must include every repeated reader, propagation, and gate loss exactly once. If a reader is traversed only once, put its transmission outside the power N. State-dependent filtering requires a quantum channel rather than one scalar number.

$$P_{\mathrm{out}}(t) \simeq \eta e^{-\kappa t}, \qquad \kappa = -\frac{\ln(T \prod_j R_j)}{\tau_{\mathrm{rt}}}, \qquad t_e = \kappa^{-1}. \tag{13}$$

This exponential reproduces the discrete model at integer circuits. In the illustrative case the survival e-folding time is about 2.668 μs. Improving a readout while reducing this survival may decrease total useful events per incident photon.

$$E_\gamma = \frac{hc}{\lambda}, \qquad E_{\mathrm{in}} + E_{\mathrm{probe}} + E_{\mathrm{control}} = E_{\mathrm{out}} + E_{\mathrm{dissipated}} + \Delta E_{\mathrm{stored}}. \tag{14}$$

Table 4. Hypothetical values: four mirrors, each $R_j = 0.9999$; $T = 1$; input and output transmission each 0.99. $T = 1$ intentionally excludes reader and switch loss, so the table is an optimistic sensitivity case, not an achievable quantum-reader prediction.

Circuits	Hold delay (μs)	Ideal survival
1	0.001067	0.979708
100	0.106741	0.941668
1000	1.067405	0.656968
10000	10.674051	0.017948

At 850 nm a photon carries approximately 2.34×10^{-19} J. A probe, switching field, and electronics use external energy. An amplified electrical record is not powered solely by that photon. Pump-generated photons and probe leakage must be separated from transmission of the signal excitation.

6 Polarization, phase, and mode preservation

A mirror can preserve photon number while rotating polarization or coupling it to frequency and direction. In a chosen local basis, an idealized mirror has complex amplitude coefficients for s and p polarization. Coordinate rotations between baffles also enter the round-trip Jones matrix.

$$J_j(\omega) = \begin{pmatrix} r_{s,j}(\omega) & 0 \\ 0 & r_{p,j}(\omega) \end{pmatrix}, \qquad J_{\mathrm{rt}}(\omega) = J_{n_{\mathrm{mir}}}(\omega) \cdots J_1(\omega). \tag{15}$$

Each factor is understood in a consistent transported basis; otherwise basis-change matrices must be inserted. Exact polarization preservation for all polarization inputs requires the total transformation, including switches, to be a scalar times the identity, or a known fixed unitary that can be compensated without knowing the input state.

$$J_{\mathrm{rt}}^{N}(\omega) = g(\omega) e^{i\phi(\omega)} I \quad \text{for uncompensated universal preservation.} \tag{16}$$

The scalar amplitude $g(\omega)$ can still filter the wave packet. A linear spectral phase adds a common delay; curvature in spectral phase distorts a pulse. Differential phase between polarization components creates an additional encoding error. Here β_2 is total transfer-phase curvature in s^2, not per-length dispersion.

$$\phi(\omega) \simeq \phi_0 + \tau_g(\omega - \omega_0) + \tfrac{1}{2}\beta_2(\omega - \omega_0)^2. \tag{17}$$

Time-bin encoding requires coherence between temporally separated components and a gate schedule that treats both correctly. A presence reader capable of distinguishing early from late can destroy time-bin coherence even if it is insensitive to polarization. Preservation claims must identify which degrees of freedom carry the qubit.

Reader environment

A baffle can exchange momentum with a photon. If its unobserved recoil or internal excitation becomes correlated with the encoding, tracing out that state affects coherence. A measurable

mechanical response does not automatically disclose polarization or a cryptographic bit. The same signal-reader channel analysis used below covers optical, electronic, atomic, and mechanical records.

$$F_{\mathrm{cond}} = \langle\psi|\rho_{\mathrm{out,cond}}|\psi\rangle, \qquad P_{\mathrm{surv}}(\psi) = \mathrm{Tr}\,\mathcal{E}_{\mathrm{survive}}(|\psi\rangle\langle\psi|). \tag{18}$$

Report conditional fidelity and the survival yield $P_{\mathrm{surv}}(\psi)$ for each input state. For entangled inputs, test preservation of joint correlations with the reference; local photon survival alone is insufficient.

7 Readers behind the mirrored walls

The reader must receive a signal-dependent physical influence. A perfectly opaque, ideal mirror with no monitored internal coupling supplies no optical field to a sensor behind it. A partially transmitting coating, evanescent interaction, cavity coupling, or monitored material response supplies a candidate interaction; each has a different loss and disturbance model.

Classical optical tap

A partial reflector sends a fraction ϵ_{tap} to a photodetector. For an ideal single-photon input and a vacuum tap port, a lossless splitter produces

$$|1\rangle_s|0\rangle_r \longmapsto \sqrt{1-\epsilon_{\mathrm{tap}}}\,|1\rangle_s|0\rangle_r + e^{i\varphi}\sqrt{\epsilon_{\mathrm{tap}}}\,|0\rangle_s|1\rangle_r. \tag{19}$$

An absorbing click in the reader branch removes that photon from the onward branch in that event. For a bright intensity-modulated beam, many photons can remain to carry a classical message, which makes the same arrangement a useful classical sampling instrument.

$$P_{\mathrm{remain}} = (1-\epsilon_{\mathrm{tap}})^K, \qquad P_{\mathrm{absorbed}} = 1-(1-\epsilon_{\mathrm{tap}})^K. \tag{20}$$

These complementary probabilities assume an ideal detector, no other losses, and K identical encounters. They count the fate of one photon. With less than unit detector efficiency, some removed photons produce no registered click, so measured readout probability is smaller.

$$\Phi_r = \epsilon_{\mathrm{tap}}\Phi, \qquad I_{\mathrm{photo}} = \mathcal{R}\,\Phi_r, \qquad \mathcal{S}_I = 2q_e I_{\mathrm{photo}}. \tag{21}$$

Here optical power Φ is in watts, responsivity $\mathcal{R}$ is in A/W, and $\mathcal{S}_I$ is the one-sided shot-noise current spectral density in A^2/Hz for an ideal Poisson photodetection model. Electronics noise, dark current, bandwidth, and excess optical noise add to the measurement budget.

Material response

Radiation-pressure readout or a probe phase shift can register another observable without direct absorption of the signal. It still needs a specified interaction strength, coupling bandwidth, temperature or probe noise model, and a way to collect the record. Merely locating electronics behind diagonal baffles does not supply those specifications.

8 Nondestructive presence reader

Nondestructive photon detection is a real experimental capability in specialized systems. Reiserer, Ritter, and Rempe demonstrated optical-photon detection using an atom coupled to a resonator [1]. Niemietz and colleagues demonstrated photonic-qubit presence detection with an atom and two resonators [2]. These results support a presence-sensing research track; they do not demonstrate this proposed device.

A useful effective model couples signal photon number to a separate probe photon number. It is a model of an interaction, not a property assumed for ordinary prism glass or a mirror coating.

$$H_{\mathrm{int}} = \hbar\chi\,\hat{n}_s\hat{n}_p, \qquad U(t) = \exp(-i\chi t\hat{n}_s\hat{n}_p). \tag{22}$$

$$|n\rangle_s|\alpha\rangle_p \longmapsto |n\rangle_s|\alpha e^{-in\chi t}\rangle_p. \tag{23}$$

For vacuum versus one signal photon, the magnitude of the two coherent-probe overlaps is

$$|\langle\alpha|\alpha e^{-i\phi}\rangle| = \exp[-|\alpha|^2(1-\cos\phi)], \qquad \phi = \chi t. \tag{24}$$

A stronger phase separation can improve ideal presence discrimination. Real nonlinear media add noise, finite bandwidth, absorption, and multimode effects. Causal response noise can invalidate simple traveling-wave Kerr gate models [7]; that result is not a prohibition on every cavity-based QND instrument. No usable χ is inferred from prism glass or an uncharacterized coating.

$$\hat{n}_s = \hat{n}_H + \hat{n}_V, \qquad U(|\psi\rangle_{\mathrm{pol}}|1\rangle_s|\alpha\rangle_p) = |\psi\rangle_{\mathrm{pol}}|1\rangle_s|\alpha e^{-i\phi}\rangle_p. \tag{25}$$

The photon-number ket denotes the fixed one-photon sector, not an additional independently populated optical mode. The last equation requires equal coupling to both polarizations and a common temporal-spatial mode. It shows why the probe can register occupancy without learning the polarization bit. If the coupling distinguishes the encoded components, this separation fails.

For an early/late time-bin qubit, two time-resolved probe records can reveal the bin and dephase that encoding. A proposed presence reader must be demonstrably insensitive to every degree of freedom whose coherence must survive. For this reason the quantum module remains a laboratory subsystem specification, not an interchangeable backing plate for a large mirror.

9 Reading a specified quantum observable

A general readout is represented by a quantum instrument with outcomes y and Kraus operators G_{yk}. Each recorded outcome has a corresponding conditional state update; summing over the record gives the output seen by a receiver who cannot access the reader.

$$\mathcal{I}_y(\rho) = \sum_k G_{yk}\rho G_{yk}^\dagger, \qquad p(y) = \mathrm{Tr}\,\mathcal{I}_y(\rho), \qquad \sum_{y,k} G_{yk}^\dagger G_{yk} = I. \tag{26}$$

Loss outcomes are included in this instrument. If ρ is polarization only, this description must be enlarged to include vacuum and any other modes whose leakage affects the result. Conditional normalization can hide state-dependent losses if they are not reported separately.

Ideal basis reader

Define Z-basis bit states as $|H\rangle$ and $|V\rangle$. A nondestructive basis reader can ideally implement

$$U|H\rangle|m\rangle = |H\rangle|m_H\rangle, \qquad U|V\rangle|m\rangle = |V\rangle|m_V\rangle. \tag{27}$$

It preserves these two eigenstates while recording them. For a superposition, linearity gives an entangled signal-reader state. With overlap S between the two normalized reader states, the receiver sees

$$\rho'_P = \begin{pmatrix} |\alpha|^2 & \alpha\beta^* S \\ \alpha^*\beta S^* & |\beta|^2 \end{pmatrix}, \qquad S = \langle m_V|m_H\rangle. \tag{28}$$

Orthogonal records give $S = 0$ and remove off-diagonal coherence. Identical records give $S = 1$ and preserve it. A known common phase can be compensated; a distinguishable record cannot be removed merely by correcting a phase.

$$|+\rangle = (|H\rangle + |V\rangle)/\sqrt{2}, \qquad Q_X = \frac{1 - \operatorname{Re} S}{2}. \tag{29}$$

A complete Z reader with orthogonal records produces a 50 percent X-basis error for this input, despite the photon surviving in the ideal model. The result explains why photon detection, photon survival, chosen-basis bit reading, and preservation of arbitrary encoding cannot be treated as equivalent claims.

10 Exact preservation and retained information

Proposition under the stated assumptions

Suppose a deterministic device receives either of two nonorthogonal pure states, uses a fixed initial apparatus state, returns each signal state exactly unchanged, and retains a record in any part of the apparatus. The record cannot distinguish the two inputs. A mixed initial apparatus can be purified by including its environment, so the proof below covers it as well.

$$U|\psi\rangle|m\rangle = |\psi\rangle|m_\psi\rangle, \qquad U|\varphi\rangle|m\rangle = |\varphi\rangle|m_\varphi\rangle. \tag{30}$$

Because each surviving output is pure and unchanged, it is uncorrelated with the rest of the final system for that input. Unitary evolution preserves inner products:

$$\langle\psi|\varphi\rangle = \langle\psi|\varphi\rangle\langle m_\psi|m_\varphi\rangle. \tag{31}$$

$$\langle\psi|\varphi\rangle \neq 0 \quad \Longrightarrow \quad \langle m_\psi|m_\varphi\rangle = 1. \tag{32}$$

The normalized records are therefore identical and no later measurement on them reveals which input occurred. This is the standard information-disturbance argument used in the security proof of Bennett, Brassard, and Mermin [3] and analysed quantitatively by Fuchs [4]. A fixed reversible signal transformation can be undone before applying the proof and offers no exception.

Why more reflections do not change the result

$$U_{\text{total}} = U_{\text{release}} U_N \cdots U_2 U_1 U_{\text{capture}}. \tag{33}$$

All mirror interactions, coherent controls, records, and reset environments belong in the total evolution. If every possible input exits exactly intact, the same overlap calculation applies to that total operation. The argument does not depend on whether the reader is optical, electronic, atomic, or mechanical.

For BB84, $|H\rangle$ and $|V\rangle$ are orthogonal, but each overlaps the diagonal states. Those connections force identical apparatus records throughout the four-state alphabet when all four states are preserved exactly. Reading a known orthogonal alphabet is a different case.

Limits of the proposition

The result excludes the universal deterministic exact claim. It does not prove that every attempted interception is identified, bound every approximate attack, or rule out information in a side channel. Probabilistic successes, discarded trials, finite tolerances, and multiphoton pulses require explicit yield and protocol analysis. No claim of an alarm on every event is made.

11 Repeated weak reading with explicit calculations

Consider repeated Z-basis interactions with fresh initially independent probes. Let each pair of conditional probe states have the same real overlap $s \in [0, 1]$. N counts circuits and K counts reader encounters; with r enabled stations per circuit, $K = rN$. Fresh independent probes at every encounter give the product

$$S_K = s^K, \qquad Q_X(K) = \frac{1 - s^K}{2}, \qquad K = rN. \tag{34}$$

For equally likely H and V inputs, the optimal discrimination probability for the two pure joint probe states is the Helstrom expression. This ideal joint measurement may require capabilities beyond a simple detector behind each wall.

$$P_{\text{guess}}(K) = \frac{1 + \sqrt{1 - s^{2K}}}{2}, \qquad (2P_{\text{guess}} - 1)^2 + (1 - 2Q_X)^2 = 1. \tag{35}$$

Table 5. Ideal selected-basis model with $s = 0.999$ and no photon loss. As records become easier to distinguish, complementary-basis coherence decreases. These are raw H/V discrimination probabilities, not probabilities of recovering a distilled cryptographic key.

Interactions	Probe overlap	X error percent	Z guess percent
1	0.999000	0.0500	52.2355
10	0.990045	0.4978	57.0376
100	0.904792	4.7604	71.2927
1000	0.367695	31.6152	96.4973

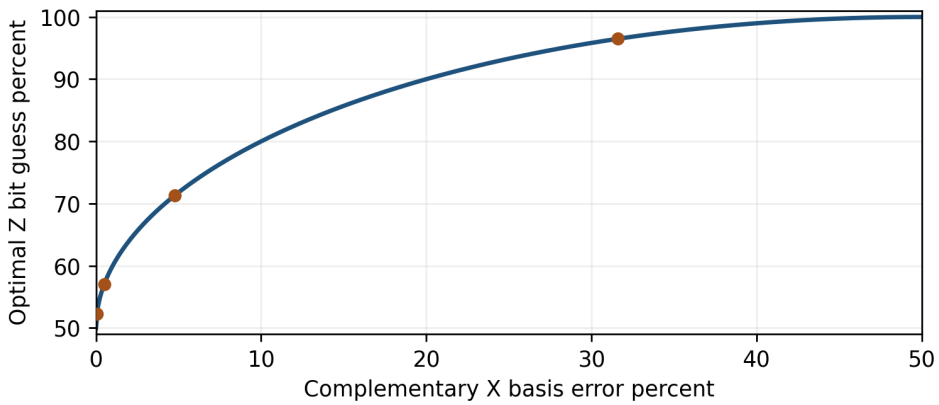


Figure 2. Exact numerical relation for the chosen ideal probe model. It is not a universal optimal attack bound for all protocols.

12 Reversal, erasure, and cryptographic interpretation

If the same probe remains coherent over all passes, the independent-probe product model need not apply. For conditional reader evolutions V_H and V_V, the overlap instead depends on their full accumulated dynamics:

$$S_K = \langle m|(V_V^\dagger)^K V_H^K|m\rangle. \tag{36}$$

Interference, saturation, revivals, or deliberate reversal can therefore change the dependence on K. A return of the overlap to unity restores coherence in this ideal model, but also makes the final distinguishing probe records identical. Reusing a probe does not evade the total-evolution proposition.

Undoing the interaction

$$U^{-1}(\alpha|H\rangle|m_H\rangle + \beta|V\rangle|m_V\rangle) = (\alpha|H\rangle + \beta|V\rangle)|m\rangle. \tag{37}$$

This coherent reversal restores the signal while removing the distinguishable record. If a second memory already copied the outcome, that memory remains part of the environment. Reversing only the local mirror does not erase the external record. Ordinary deletion of a computer file is not coherent quantum erasure.

Raw data and a final secret key

A quantum key distribution system tests relevant statistics, reconciles bits over an authenticated classical channel, and applies privacy amplification using a bound on possible adversary knowledge. Photon collection or partial raw-bit information does not by itself establish knowledge of the final secret key. Decoy-state methods address source imperfections from multiphoton components under defined assumptions [5].

Let Z_{raw} be the raw-key register, E the adversary's quantum side information, C the public transcript, and F an independent uniform seed selecting a two-universal hash that maps Z_{raw} to the final key register W. For this classical-quantum state the unsmoothed leftover-hash lemma gives the

conservative sufficient rule below [6]. A factor-two triangle-inequality conversion to the actual side-information marginal is included. Here d_{tr} is half the trace norm and u_W is the uniform key state.

$$\begin{aligned} d_{\mathrm{tr}}(\rho_{WFEC}, u_W \otimes \rho_{FEC}) &\le \sqrt{2^{\ell - H_{\min}(Z_{\mathrm{raw}}|EC)}}, \\ \ell &\le \left\lfloor H_{\mathrm{lb}} - 2\log_2 \frac{1}{\epsilon_{\mathrm{PA}}} \right\rfloor, \quad H_{\mathrm{lb}} \le H_{\min}(Z_{\mathrm{raw}}|EC). \end{aligned} \tag{38}$$

Use a certified lower bound H_{lb} on the conditional min-entropy after disclosure of C and choose $0 < \epsilon_{\mathrm{PA}} < 1/2$. The stated nonnegative integer length guarantees secrecy distance at most ϵ_{PA} for the hashing step; a negative bound means extract no key. This conservative unsmoothed rule is not an optimality bound. Transcript leakage is already included in conditioning on C and must not be subtracted twice. Parameter-estimation failure, authentication, and correctness need separate accounting; no link-specific H_{lb} is estimated here.

Added delay, loss, mode mismatch, and basis errors may each be detectable statistically. None alone guarantees attribution to an interceptor. Conversely, low measured bit error alone is not a complete security proof.

13 Experimental program and estimators

Begin with a controlled source and receiver on an optical bench. Log prepared states, gate settings, trials, reader outcomes, output detections, and timing. Keep vacuum trials and unsuccessful release events in the record. Use an independent reference path and randomized probe-on and probe-off blocks.

Table 6. Staged experimental program.

Stage	Question and required observation
Routing	Does the finite folded path have the predicted exit and delay?
Classical sampling	Does calibrated tap power predict measured readout?
Quantum baseline	What are survival and state errors with the reader disabled?
Presence sensing	Does the probe predict occupancy above vacuum controls?
Basis reading	How do useful bit records change complementary-basis errors?
Open-air evaluation	How do background and beam wander alter the bench result?

$$\widehat{P}_{\mathrm{out}} = \frac{n_{\mathrm{out}}}{n_{\mathrm{trials}}}, \qquad \widehat{Q}_X = \frac{n_{X,\mathrm{wrong}}}{n_{X,\mathrm{accepted}}}, \qquad \widehat{p}_{\mathrm{false}} = \frac{n_{\mathrm{reader\ click,vac}}}{n_{\mathrm{vac}}}. \tag{39}$$

These are operational detection probabilities. Inferring device-only survival requires independent calibration of source preparation, detector efficiency, dark counts, and heralding. Publish raw and corrected values and the correction model. Dark-count subtraction alone can be unreliable when backgrounds drift or events overlap.

$$\widehat{p} \pm z\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}} \quad \text{is only a large-sample approximation.} \tag{40}$$

Use Wilson or exact binomial intervals for independent count data, especially near zero or one. If no failures are observed in n independent trials, a one-sided 95 percent upper bound is approximately $3/n$; zero observed failures does not establish zero failure probability. Correlated drift requires block analysis or another appropriate uncertainty model.

Prepare H, V, diagonal, antidiagonal, and circular polarization states to characterize the channel, then use held-out randomized states for validation. Test entangled inputs when preservation of external entanglement is claimed. For timing encodings, use early/late superpositions as well as isolated pulses. Compare matched resources: same incident flux, bandwidth, probe energy, and total interaction opportunity.

14 Decision criteria and implementation boundaries

The prototype must separate four outcomes: whether the photon arrived, whether it survived, what the reader learned, and what state exited. A high conditional fidelity can coexist with severe loss or preferential removal of difficult states. The acceptance rule must therefore include state-resolved yield and readout performance.

$$P_{\text{useful}} = P_{\text{surv}}\, P_{\text{correct record}|\text{surv}}, \qquad I(B:Y) = \sum_{b,y} p(b,y) \log_2 \frac{p(b,y)}{p(b)p(y)}. \tag{41}$$

The first expression is a bookkeeping probability, not a security metric. The mutual information between the input-bit variable B and the record Y measures correlations for a defined ensemble and recorded measurement; it does not identify the information available to an arbitrary quantum adversary or the final key entropy.

Table 7. Acceptance criteria to be quantified before data collection.

Requirement	Proposed decision rule
Ray routing	All nominal rays and full beam envelopes clear measured apertures
Timing	Measured gate transition and jitter fit the return schedule
Presence claim	Probe outcomes separate photon and vacuum controls with intervals
State claim	Full input set meets a predeclared fidelity and yield criterion
Bit-reading claim	Held-out records beat the prior guess rate and back action is reported
Long-term hold	Stable transverse mode and measured loss replace plane-ray assumptions

Numerical fidelity, throughput, and error thresholds must be set from a specific intended application before data collection. This paper does not invent a universal acceptable error rate or a universal QKD alarm threshold.

Functional bill of materials

A routing prototype needs a translation-mounted frame, four adjustable mirrors, an enclosed optical path, input and output mode-matching optics, a synchronized pulsed source, optical gating or a fixed-delay alternative, a reference channel, power and timing instruments, and a detector. The classical reader adds a calibrated tap and photodetector. A quantum experiment adds characterized photon preparation, analyzers, low-background photon detectors, time tagging, and a chosen atom-cavity

or other experimentally supported interaction module.

Part numbers, costs, mirror curvature, switch bandwidth, wavelength, and a fabrication drawing remain open because their requirements depend on the source and chosen reader. The six-inch frame concept cannot determine a microscopic coupling strength, gate technology, or achievable readout fidelity.

15 Results, limitations, and research disposition

The proposed 80 mm square centerline fits its nominal mirror centers inside a six-inch front-frame envelope, with limited radial allowance. Its 0.320 m optical loop gives a 1.067 ns transit in the stated approximation. This establishes a geometric starting point and exposes the stringent gate timing; it does not establish a complete stable cavity or packaged device.

The loss calculations quantify the consequence of repeated reflections. Even with 99.99 percent power reflectivity at every mirror and no other circuit loss, 10 000 circuits retain only about 1.79 percent unconditional transmission with the chosen fixed couplings. Real reader and switch losses lower that value. The illustrative model is deliberately transparent about those omitted losses.

The wall-reader analysis identifies three distinct mechanisms. An absorbing tap samples classical light but removes a detected single photon. A nondestructive presence interaction can preserve an unmeasured encoding when its coupling is insensitive to that encoding. A basis reader can preserve eigenstates while disturbing their superpositions. These distinctions apply irrespective of the baffle angle or the electronics location.

The exact-preservation proof includes capture, circulation, readout, release, and every retained record. It rules out deterministic extraction of distinguishing information for a nonorthogonal alphabet with all signal states returned exactly unchanged. The repeated weak-probe calculation gives a concrete tradeoff rather than a blanket assertion that every measurement destroys a photon.

Research decision

Proceeding with a controlled routing and classical-readout prototype is technically meaningful. A quantum version requires selecting and independently characterizing the physical interaction first. The requested universal intact unknown-key reader is not established by this geometry and is incompatible with the exact assumptions proved here. Device imperfections or side channels require a separate implementation-specific study; they cannot be assumed as a completed capability.

Completion record

Completed for this paper: analytical derivations, a dimensioned chief-ray example, loss and timing tables, a readout model, reproducible numerical verification programs, a document consistency checker, and an experimental evaluation plan. Not performed: electromagnetic or mechanical finite-element simulation, microscopic material design, component selection, fabrication, quantum-state measurements, independent laboratory replication, external peer review, or prior-art assessment.

Appendix A Numerical verification and reproducibility

The baseline verify_models.py program uses NumPy to calculate the square-ring normals and reflected directions, propagation time, mirror-loss table, Gaussian example, and selected-basis probe model. It constructs the joint signal-probe state explicitly for representative overlaps and compares its partial trace with the derived density matrix.

The extended review_checks.py program tests arbitrary complex input and probe states, coherent probe reuse, a physical loss channel, and the extraction-bound arithmetic. The baseline program checks the Helstrom trace-distance expression against the two probe density matrices, verifies normalization and positivity, and verifies the ideal tradeoff identity. It also checks the passive-tap survival complement. The check_document.py program, new in V004, recomputes every quoted number in the prose and confirms that each cross-reference resolves. These checks test the mathematical implementation; they are not physical experiments or evidence of hardware performance.

$$|m_H\rangle = |0\rangle, \qquad |m_V\rangle = S|0\rangle + \sqrt{1-S^2}|1\rangle, \qquad 0 \le S \le 1. \tag{42}$$

$$|\Omega\rangle = \frac{|H\rangle|m_H\rangle + |V\rangle|m_V\rangle}{\sqrt{2}}, \qquad \rho_P = \mathrm{Tr}_M\, |\Omega\rangle\langle\Omega|. \tag{43}$$

For numerical evaluation, the trace distance between the equally likely pure reader states is $\sqrt{1-S^2}$. The code diagonalizes their density-matrix difference independently and compares the result. Endpoint $S = 1$ gives no distinguishing record and zero X error; endpoint $S = 0$ gives perfect basis discrimination and 50 percent X error.

Package contents

Table 8. Package contents.

File	Purpose
Prism_Gate_White_Paper_V004.pdf	Typeset reading copy (LaTeX)
Prism_Gate_White_Paper_V004.docx	Editable Word version with native, editable equations
whitepaper_content.py, content.json	Single source for text, LaTeX equations, tables, and labels
verify_models.py, review_checks.py	Baseline and extended numerical verification
check_document.py	Recomputes quoted numbers; checks references and numbering
build_pdf.py, build_docx.py	Document generation from the single source
*_results.json, review_findings.json	Recorded check outputs and review findings
manifest.json, prior/	Integrity hashes; the unchanged V003 package (which nests V002 and V001)

Appendix B References and provenance

[1] Reiserer A, Ritter S, Rempe G. Nondestructive detection of an optical photon. Science 342, 1349–1351 (2013). DOI 10.1126/science.1246164.

[2] Niemietz D, Farrera P, Langenfeld S, Rempe G. Nondestructive detection of photonic qubits. Nature 591, 570–574 (2021). DOI 10.1038/s41586-021-03290-z. arXiv:2103.14677.

[3] Bennett CH, Brassard G, Mermin ND. Quantum cryptography without Bell's theorem. Physical Review Letters 68, 557–559 (1992). DOI 10.1103/PhysRevLett.68.557.

[4] Fuchs CA. Information gain vs. state disturbance in quantum theory. Conference talk preprint, arXiv:quant-ph/9605014 (1996).

[5] Trushechkin AS, Kiktenko EO, Kronberg DA, Fedorov AK. Security of the decoy state method for quantum key distribution. Physics–Uspekhi 64(1), 88–102 (2021). DOI 10.3367/UFNe.2020.11.038882.

[6] Tomamichel M, Schaffner C, Smith A, Renner R. Leftover hashing against quantum side information. IEEE Transactions on Information Theory 57, 5524–5535 (2011). DOI 10.1109/TIT.2011.2158473. arXiv:1002.2436.

[7] Shapiro JH. Single-photon Kerr nonlinearities do not help quantum computation. Physical Review A 73, 062305 (2006). DOI 10.1103/PhysRevA.73.062305.

Relationship to prior documents

The earlier Prism Gate Diagonal Baffle Photon Reader Concept and Experimental White Paper, version 1.0, supplies the six-inch slide-in frame, diagonal baffle arrangement, wall-reader placement, and proposed development stages. The V001 Design and Analysis package supplies an earlier ring-buffer analysis with a 1 m example. This paper uses a 0.320 m dimensioned example; differences in the timing tables follow from that changed length, not a recovered hardware measurement.

References [1] and [2] support the existence of specialized nondestructive optical detection. References [3] and [4] support the information-disturbance principle. Reference [5] supports the role of decoy-state analysis for imperfect sources, [6] supplies the privacy-amplification bound, and [7] the finite-response limit on traveling-wave Kerr gates. The geometry, numerical assumptions, state-evolution examples, and proposed tests in this document are analytical work for this concept. They are not presented as experimental results from those papers.

Attribution and claim boundaries

Levi Burd supplied the prism gate, reciprocating storage, diagonal baffles, readers behind mirrored walls, and controlled-release concept. ChatGPT assisted in technical development, calculations, and preparation of V001 to V003. Claude (Anthropic) independently cross-checked V003 and prepared the V004 revision. Established optical and quantum principles remain established work; no patentability, novelty, physical construction, or independent scientific validation is claimed. A published or generated paper is not itself a demonstration of the proposed instrument.

Appendix C.1 Technical review findings (V003)

Internal technical review by ChatGPT, October 27, 2026. This is a self-review by the authoring model, not external peer review, experimental validation, or an independent laboratory assessment. Scope: every numbered equation, the prose and tables, cited source support, numerical implementation, operational assumptions, and rendered presentation.

Overall disposition

Retain as a conditional theoretical concept and bench research proposal. The algebra supporting no information without disturbance and the baseline numerical tables survived review. The package is not ready for fabrication as an integrated quantum reader. The sensor interaction, stable optical mode, timing schedule, and source-receiver specification remain open.

Table 9. V003 internal review findings (equation numbers updated to this edition).

Finding	Severity and disposition
R1 Reader integration	High. Measured reader response and loss are not supplied; build readiness remains blocked.
R2 Timing model	Moderate, corrected. Equation 10 includes resonant module group delay and its domain.
R3 Interaction count	Moderate, corrected. Equations 34 to 36 use K encounters with $K = rN$; N counts circuits.
R4 Key-length shorthand	Moderate, corrected. Equation 38 states a sufficient unsmoothed hashing rule with assumptions.
R5 Test coverage	Moderate, expanded. Complex-state, coherent-reuse and loss-channel checks added.
R6 Physical coupling	High, unresolved design requirement. Mirrors need a specified sensor interaction.
R7 Wording	Minor, clarified. Positive frequencies, fixed photon sector and scope of capture.
R8 Source support	Moderate, corrected. Primary support for privacy amplification and nonlinear-response limits.

R1 and R6 were already acknowledged as unknowns in V002; the review made their blocking consequences explicit. None of the corrections establishes recovery of an unknown quantum key with no disturbance.

Appendix C.2 Reader integration feasibility

The ideal fresh-probe model requires a separate effectively independent record for each of K encounters. A repeatedly reused atom or probe does not meet that assumption merely because the photon reflects again. Readout and reset can be much slower than the bare loop transit; parallel probes or a coherent shared-probe model would require a different architecture.

Reference [2] reports atomic-state detection by cavity-enhanced fluorescence lasting typically 7.5 μs and a photon survival probability of (31 ± 1) percent. Both values were confirmed against the source for V004. They are reported parameters of that experiment, not specifications of this proposal. Readout duration must not be confused with photon interaction time or a universal reset limit. The source does not demonstrate a fresh reader every 1.067 ns.

$$\frac{7.5\ \mu\text{s}}{1.067405\ \text{ns}} \simeq 7026.4 \quad \text{bare-loop circuits per reported readout interval.} \tag{44}$$

As an explicit sensitivity calculation only, assume each circuit traverses a hypothetical independent

module with survival 0.31, in addition to the optimistic mirrors. Ten such traversals leave about eight parts per million. This extrapolation does not predict the cited device under repeated use; it shows why reader loss cannot be set to unity when claiming integrated performance.

$$P_{\mathrm{out}}(10) = 0.99^2(0.9999^4 \times 0.31)^{10} \simeq 8.0011 \times 10^{-6}. \tag{45}$$

A coherent presence probe that flips once per pass supplies a simple counterexample to the fresh-record assumption. For vacuum the probe stays in its initial state; for one photon its evolution is X^K. Odd encounters can distinguish presence, while even encounters restore the initial probe state. More encounters therefore need not yield more retained information.

$$|m\rangle = |0\rangle, \qquad \langle 0|X^K|0\rangle = \begin{cases} 0, & K \text{ odd}, \\ 1, & K \text{ even}. \end{cases} \tag{46}$$

The loop needs a measured component transfer function, a compatible pulse bandwidth, and a stable spatial mode. Group delay alone cannot describe severe distortion or overlapping pulses. The proposed two-dimensional centerline also leaves injection, extraction, internal supports and three-dimensional access for sensors unresolved.

Appendix C.3 Extended verification and remaining tests

The reproducible review uses a fixed random seed. It evaluates 500 normalized complex polarization states and pairs of normalized complex probe states in a three-dimensional probe space. For each case it constructs the joint state, traces out the probe, and compares every output matrix entry with Equation 28. It independently diagonalizes the probe-state difference to verify the equal-prior Helstrom discrimination result.

A second 500-case group constructs the corresponding dephasing-channel Choi matrix and checks positivity and the trace-preserving partial trace. A 64-step coherent-probe test verifies oscillatory record overlap and the bit-flip parity example. A 101-case attenuation-channel sweep includes a vacuum output and tests trace preservation and the separation of conditional fidelity from survival.

Additional checks recompute the integration sensitivity example, compare source tables with numerical output, and verify the hashing-length rule over a parameter grid. These checks are reproducibility and consistency tests. They are not an independent proof of every physical assumption or a substitute for sensor characterization.

Dimensional and source audit

The ray directions are dimensionless; Gaussian radii and path coordinates have length units; group delay and χ^{-1} have time units; reflectivities and overlaps are dimensionless; the shot-noise density $\mathcal{S}_I$ is in A^2/Hz. In Equation 17, β_2 denotes total transfer-phase curvature with units s^2, not the per-length fiber dispersion parameter. In Equation 21 the spectral density is one-sided. A general birefringent path still needs explicit basis rotations.

The paper references actual nondestructive presence experiments and a standard information-disturbance result. Neither supplies the proposed large-aperture wall material or a fabrication recipe. The ideal Kerr Hamiltonian remains an effective model, with finite-response caveats explicitly cited. The hashing result is a sufficient secrecy bound, not a link-specific rate. For the source theorem

optimized over a side-information state, tracing out the key bounds the marginal distance by the same δ; the triangle inequality then bounds the actual-marginal distance by 2δ. This explains the conservative constant in Equation 38.

Tests that still require a laboratory

Measure per-station coupling, bandwidth, loss, polarization response and timing; establish the cavity mode; characterize gate rise time and jitter; test photon and vacuum controls; perform state-resolved tomography and entanglement-preservation tests; and report every loss and rejected event. Only these measurements can determine whether a chosen reader and a particular optical buffer can operate together.

Appendix D Independent cross-check of V003 (V004 changes)

V003 was cross-checked by Claude (Anthropic), a model that did not write V001 to V003. This removes the self-grading limitation of Appendix C.1 for the checks listed here. It is still not external human peer review and not laboratory validation.

What was checked

Every numeric value quoted in the prose and tables was recomputed from first principles in a separate calculation: ring geometry and clearance, loop transit, Table 4, the e-folding time, Gaussian beam values, clipping fraction, photon energy, Table 5, the readout ratio, and the integration sensitivity case. All matched to the stated precision. Both verification programs were re-run and reproduced their recorded outputs exactly. Each derivation was checked by hand, including the coherent-state overlap in Equation 24, the dephasing in Equation 28, the X error in Equation 29, the Helstrom identity in Equation 35, the exact-preservation proposition, and the factor-two constant in Equation 38. Bibliographic data for all references, and the two numerical parameters taken from reference [2], were checked against the publishers or arXiv.

No errors of substance were found

The corrections below are presentational, notational, and bibliographic. No equation in V003 was mathematically wrong, and no conclusion changes.

Table 10. V004 cross-check findings.

Finding	Severity and disposition in V004
R9 Symbol collisions	Minor, corrected. K had meant encounters, Kraus operators and the key register; S meant overlap and shot-noise density; Z meant a basis and the raw key; P meant a system and optical power; h meant Planck's constant and an entropy bound; m meant mirror count and apparatus state. Each symbol now has one meaning (Appendix E).
R10 Presentation	Minor, corrected. Every page footer read "Proposed architecture"; three appendices shared the label C; captions were body text. Running footers now name the section, appendices are C.1 to C.3, and tables and figures carry numbered captions.
R11 Editability	Minor, corrected. Word equations were images. They are now native, editable Word equations; the PDF is typeset directly by LaTeX.
R12 Cross-references	Minor, hardened. Equation, table and figure numbers are resolved from labels at build time, and check_document.py fails if a reference or quoted number drifts.
R13 Citations	Minor, corrected. The primary source of the Section 10 argument, Bennett–Brassard–Mermin [3], was added. Reference [4] is identified as a conference-talk preprint. Reference [2] parameters confirmed.
R14 Title claim	Minor, corrected. "Mathematical and experimental white paper" implied experiments had been performed; the subtitle now reads "experimental program".
R1, R6	Unchanged and still open. No edit to the document can close them; they require a selected reader interaction and measured loss.

Appendix E Notation

Each symbol has one meaning throughout. Subscripted quantities are listed where the subscript changes the meaning.

Table 11. Notation.

Symbol	Meaning	Where
A, B, C, D; $\mathcal{M}$	Ray-transfer matrix elements; round-trip matrix	Eq. 9
$a_\sigma^\dagger(\omega)$	Creation operator, polarization σ	Eq. 1
B, b	Input-bit random variable; its value	Eq. 41
C, E, F	Public transcript; adversary side information; hash seed	Eq. 38
D_c, L_s, L_rt	Mirror-center diameter; square side; round-trip path	Eq. 5
d_tr, u_W	Trace distance (half trace norm); uniform key state	Eq. 38
E_γ, E_in, …	Photon energy; energy-budget terms	Eq. 14
ϵ_tap, ϵ_clip, ϵ_PA	Tap fraction; clipped fraction; secrecy parameter	Eqs. 19, 8, 38
η, T, R_j	Coupling product; nonmirror transmission; mirror reflectivity	Eq. 12
F_cond, $P_\mathrm{surv}(\psi)$	Conditional fidelity; state-resolved survival yield	Eq. 18
G_{yk}, $\mathcal{I}_y$	Kraus operators; quantum instrument	Eq. 26
$g(\omega)$, $\phi(\omega)$, β_2	Scalar transfer amplitude; spectral phase; phase curvature	Eqs. 16, 17
H_min, H_lb, ℓ	Conditional min-entropy; certified lower bound; key length	Eq. 38
h, $\hbar$, c, q_e	Planck constant; reduced Planck constant; speed of light; electron charge	Eqs. 14, 22
J_j, J_rt	Jones matrices	Eq. 15
K, r, N	Reader encounters; encounters per circuit; circuits ($K = rN$)	Eq. 34
κ, t_e	Survival decay rate; e-folding time	Eq. 13
M, $\lvert m\rangle$, $\lvert m_H\rangle$	Reader system; initial and conditional reader states	Eqs. 3, 27
n_mir	Mirror stations per circuit	Eq. 12
P	Signal system (unsubscripted only)	Eq. 3
P_out, P_guess, P_useful	Output, guessing, and useful-event probabilities	Eqs. 12, 35, 41
Φ, Φ_r, $\mathcal{R}$, $\mathcal{S}_I$	Optical power; tapped power; responsivity; shot-noise density	Eq. 21
Q_X, Q_Z	Complementary-basis error rates	Eq. 29
S, S_K, s	Reader-state overlap; after K encounters; per encounter	Eqs. 28, 34
τ_rt, τ_hold, τ_g	Round-trip time; hold time; group delay	Eqs. 10, 17
W, Z_raw	Final key register; raw-key register	Eq. 38
X, Z	Measurement bases (also the Pauli X in Eq. 46)	Eqs. 29, 46
Y	Recorded reader outcome	Eq. 41
χ, $\phi = \chi t$	Effective cross-Kerr coupling; probe phase	Eqs. 22, 24